

Acknowledgment

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CHAPTER: 1

Introduction

1.1 About Company

Ascent Enterprises is a manufacturing company that specializes in the production of precision mechanical components and sheet metal assemblies for the automotive and heavy machinery industries. The company is strategically located in the Chikhali industrial corridor of Pune and boasts a team of highly skilled mechanical engineers, programmers, and technicians who utilize advanced subtractive technology to create high- quality parts.

The global and domestic manufacturing sectors are currently undergoing a massive paradigm shift, driven by the principles of intelligent automation and lean manufacturing. To thrive in this highly competitive landscape, Ascent Enterprises continuously optimizes its operations to reduce cycle times, minimize material scrap, and ensure consistent product quality.



Fig1: Precision Machining Setup at Ascent Enterprises

With a profound focus on engineering innovation and continuous process improvement, Ascent Enterprises remains a dedicated leader in its sector, bridging the gap between raw material procurement and the delivery of highly toleranced precision parts to various industrial supply chains.

1.2 Departmental Integration and Services

The company's core services include CNC turning, milling, welding, finishing, and rigorous quality inspection. Ascent Enterprises has an unwavering commitment to quality, safety, and customer satisfaction, working closely with elite clients to provide customized engineering solutions that meet exact dimensional tolerances and specific metallurgical needs.

The shop floor is meticulously organized, comprising several dedicated and interconnected departments. These include the primary Production Department, where heavy material removal occurs; the Assembly Unit, where multi-part components are constructed; the Quality Control (QC) laboratory, which acts as the ultimate gatekeeper for product standards; and the Stores/Inventory management division, which ensures the continuous supply of required raw materials and tooling inserts.

By experiencing these specific departments, this internship served as an ideal ecosystem to observe the intersection of traditional manufacturing processes and modern operational optimization. It offered a unique vantage point to study how live constraints—such as machine cycle times, progressive tool wear, material logistics, and operator ergonomics—dictate the overall efficiency and profitability of a manufacturing plant.

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Objective

- **Process Mastery:** To gain hands-on exposure to CNC turning, milling, and advanced subtractive technologies for high-precision component manufacturing.
- **Workflow Integration:** To understand the coordination between Production, Assembly, Quality Control (QC), and Inventory departments.
- **Operational Analysis:** To study how machine cycle times, tool wear, and logistics directly impact plant efficiency and profitability.

CHAPTER: 3

Methodological Details

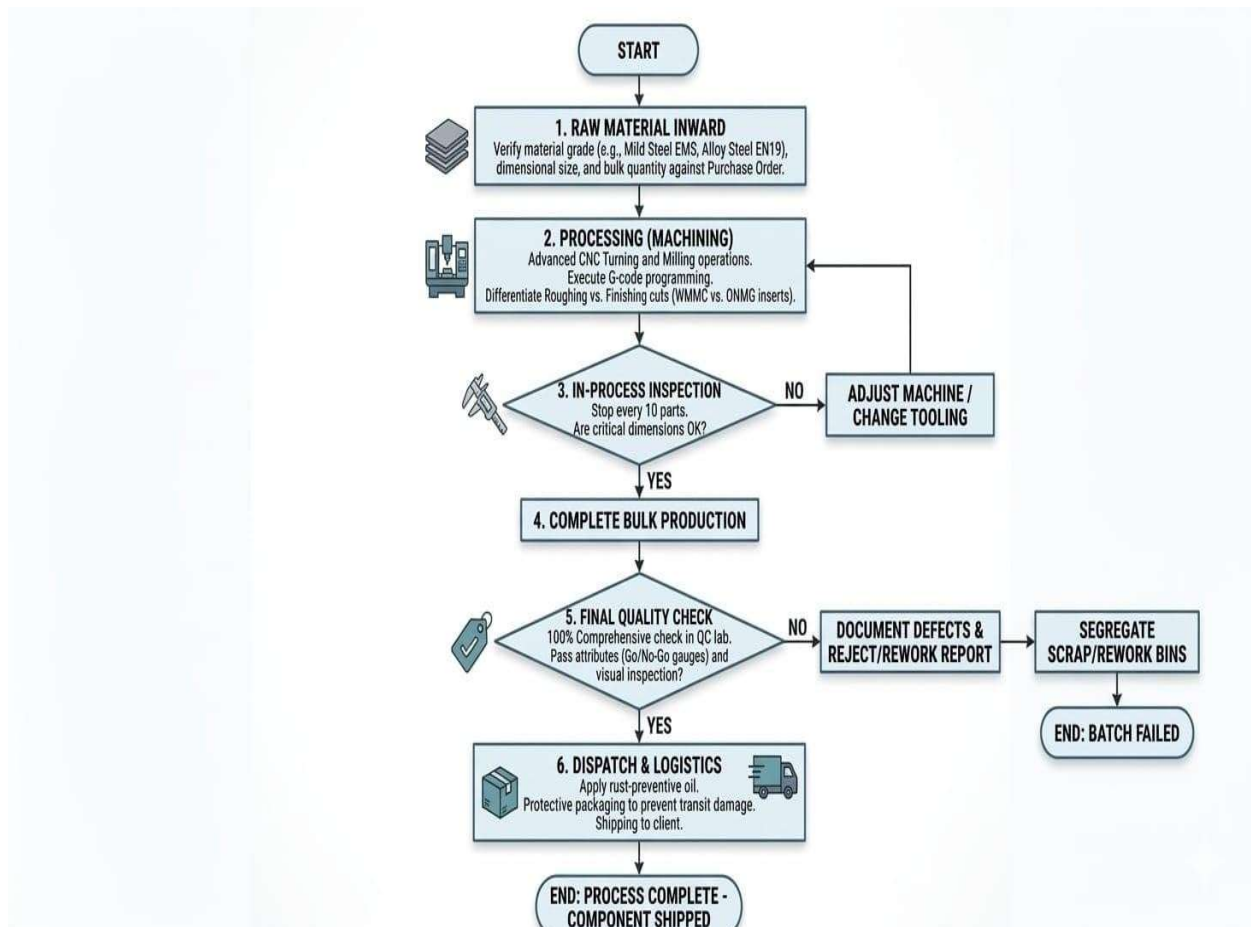
3.1 The methodological details of my internship were as follows:

- **Orientation:** The internship started with an orientation session where I was introduced to the company's policies, organizational structure, and the specific projects I would be working on. This orientation helped me get a better understanding of the professional expectations and strict safety requirements for the internship.
- **Training:** I received detailed technical training on the company's core production processes and quality control procedures. This training provided me with a foundational understanding of the precision manufacturing industry and the vital importance of dimensional accuracy.
- **Teamwork:** Throughout the internship, I worked closely with a team of professionals from different departments, such as production, quality control, and maintenance. This collaboration provided me with the opportunity to learn from experienced professionals and develop my shop floor teamwork skills.
- **Feedback and Evaluation:** I consistently received feedback on my performance from my industrial supervisor. This feedback helped me identify technical areas where I needed to improve and provided me with an opportunity to grow and develop my engineering skills.

Safety Standards (HSE) & Mandatory PPE Compliance

Industrial environments present inherent physical risks. Therefore, rigorous compliance with Personal Protective Equipment (PPE) was taught as a mandatory methodology:

- **Safety Shoes:** Heavy-duty steel-toed boots were mandatory to protect the feet against severe crush injuries from dropped raw material bars or heavy fixtures.
- **Gloves:** High-grade cut-resistant industrial gloves were utilized when operators handled razor-sharp metal chips, burrs, and freshly machined components directly from the lathe bed.



1. Raw Material Inward

The process begins with "Inward" inspection. Before any work starts, the material must be verified against the Purchase Order.

Grade Check: Ensuring the metal is correct (e.g., Mild Steel MS vs. Alloy Steel EN19). Using the wrong grade can lead to tool breakage or part failure.

Dimensional & Quantity: Checking if the bars or sheets are the right size and that the full order has arrived.

2. Processing

This is the core value-addition stage where the raw material is shaped.

Operations: Using CNC Turning (lathes) or Milling (VMC) machines.

Tooling Strategy: This stage mentions differentiating between Roughing (removing bulk material quickly) and Finishing (achieving the final smooth surface) using specific inserts like WNMG or DNMG.

3. In-Process Inspection

Rather than waiting until the end to check parts, the operator stops every 10 parts.

Decision Point: If the dimensions are within tolerance (OK), the process continues.

Correction Loop: If the dimensions are "No," the machine is adjusted or the worn-out tool is changed. This prevents producing a whole batch of defective parts.

4. Complete Bulk Production

Once the in-process checks confirm the setup is stable, the remaining quantity is machined to completion.

5. Final Quality Check

This is a 100% Comprehensive Check performed in a dedicated Quality Control (QC) lab, away from the heat and vibration of the shop floor.

Verification: Technicians use Go/No-Go gauges (to check thread or hole tolerances quickly) and visual inspection for surface scratches or burrs.

Failure Path: If a part fails, it doesn't just disappear. It is documented, rejected, and moved to Scrap/Rework Bins. This ensures defective parts never reach the customer.

6. Dispatch & Logistics

The final stage prepares the "healthy" parts for the journey to the client.

Protection: Machined steel parts can rust quickly due to humidity; therefore, rust-preventive oil is applied.

3.2 NATURE OF WORK (WEEKLY ACTIVITIES)

We were segregated into groups and were assigned different departments on a weekly basis to comprehensively study the plant. Following are the different areas studied during our internship:

Week 1: Orientation, Tool Room & Metrology Fundamentals

- **Induction and Plant Layout:** Toured the shop floor to study the highly efficient material flow layout, designed to minimize intra-plant transport time between raw material storage, CNC turning, and the dispatch area.
- **Blueprint Reading (GD&T):** Refreshed fundamental engineering concepts by learning to accurately interpret complex, multi-view 2D blueprints utilized on the floor. This included understanding Geometric Dimensioning and Tolerancing (GD&T) symbols such as concentricity, runout, and surface flatness.
- **Metrology Basics:** We studied different types of measuring instruments in the Tool Room. Gained practical experience in the calibration room, practicing measurement techniques with Vernier Calipers (0.02mm accuracy) and Micrometers (0.01mm accuracy). Moving on, we learned how specific tooling errors are corrected using secondary operations like surface grinding.

Week 2: Raw Material Department & Logistics

- **BOM Analysis and Kitting:** We here studied about how the company sends purchase orders (PO) and what the role of the invoice is in buying the raw materials. Reviewed the intricate Bill of Materials (BOM) to verify that all necessary components were available prior to assembly.
- **Material Verification:** Systematically learned how operators verify raw material grades (e.g., Mild Steel EN8, Alloy Steel) against design specifications and supplier Material Test Certificates (MTC) prior to authorizing machine loading. Learned various new terms used in the company while contacting suppliers.

Week 3: Production Workflow & CNC Machining

The production department heavily depends upon the specific requirements of the mechanical components ordered by the client. The production head coordinates with the workers on a daily basis regarding the shift targets.

- **CNC Machine Operation Analysis:** Observed live CNC turning operations. Successfully distinguished the differing programming parameters and tooling requirements for Roughing cuts (which prioritize rapid material removal with higher feed rates and larger depth of cut) versus Finishing cuts (which require specific, precise inserts, slower feeds, and higher spindle RPMs to achieve the required Ra surface finish).
- **Workholding and Clamping Dynamics:** Learned and practically observed the highly critical procedures for safe "Jaw Clamping" on the hydraulic lathe chuck. Understanding the relationship between hydraulic clamping pressure and workpiece rigidity is vital for preventing dangerous workpiece vibration (chatter), preventing part distortion, and avoiding complete part ejection during high-speed spindle rotation



Fig2: CNC lathe machine



Fig3: Vertical Machining Center(VMC)

This week heavily underscored the importance of theoretical machining parameters and how G-code/M-code logic translates into precise mechanical movements via servo motors and ball screws on the production floor.

Week 4: Quality Control (QC) & Inspection

Week four was dedicated entirely to the Quality Control laboratory and the in-line inspection stations on the shop floor. Quality assurance acts as the critical gatekeeper that ensures Ascent Enterprises maintains its reputation by strictly preventing defective components from reaching the market.

- **Comprehensive Inspection Process:** Conducted a rigorous, manual 100% Inspection process on a specialized, high-tolerance batch of 50 critical automotive components. The primary goal was to meticulously filter out manufacturing defects. Subsequently, assisted senior QC staff in evaluating off-spec dimensions, documenting physical

anomalies, and preparing the official Rejection and Rework Reports for non-conforming items.

- **Tools & Geometric Techniques:** Extensively utilized attribute gauges, specifically Go/No-Go Plug Gauges (for verifying internal hole diameters) and Snap Gauges (for verifying external shaft diameters). These essential tools allow for rapid, highly accurate "Pass/Fail" decision-making directly on the production line without requiring complex digital readouts.
- **Visual and Surface Inspection:** Performed detailed Visual Inspections under bright, specialized magnification lighting to check for critical surface anomalies. This included identifying microscopic handling scratches, tooling dents, chatter marks resulting from vibration, or uncut internal burrs that could critically interfere with the final mechanical assembly process.
- **Root Cause Outcome Analysis:** Through continuous inspection data tracking, we successfully identified that a majority of rejected defects were caused directly by progressive CNC tool insert wear (leading to oversized or undersized dimensions) or microscopic improper initial workpiece seating against the chuck face caused by fatigued manual operators.

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Scope for Future Work

4.1 The future scope of my internship is highly significant in terms of my academic and professional growth. Here are some potential future benefits that I can derive from this intensive industrial experience:

- **Improved Career Prospects:** The practical knowledge and hands-on experience gained directly on the production floor can help immensely improve my career prospects. The exposure to real-world engineering scenarios, machine constraints, and industrial timelines provides a significant competitive edge while applying for future job opportunities in manufacturing or automation.
- **Enhanced Technical Skills:** The specific skills and knowledge gained during the internship—such as interpreting GD&T blueprints, understanding CNC cutting parameters, using precision metrology tools, and analyzing cycle times—can help me build a robust technical foundation which will be invaluable in my future academic pursuits or a demanding job in the industry.
- **Application of Learned Concepts:** The practical application of the concepts learned during the internship can help me better understand the theoretical mechanical and kinematics concepts taught in the classroom. Seeing how thermal expansion affects a steel billet during turning, or how hydraulic pressure secures a part, enables me to connect the dots and have a more profound, realistic understanding of the subject matter.
- **Expanded Professional Network:** The internship experience provided me with the unique opportunity to network with experienced professionals from different departments, including plant managers, QC inspectors, and CNC programmers. These professional connections can potentially open doors to future career opportunities, technical mentorships, or industry collaborations.
- **Strengthened Soft Skills:** The internship experience has also allowed me to significantly strengthen my soft skills, such as inter-departmental teamwork, clear technical communication on a noisy shop floor, and rigorous time management. These skills are invaluable in any engineering profession, and their continuous development will benefit me throughout my entire career.

CHAPTER: 5

Discussion and Result Analysis

5.1 The result and discussion of my comprehensive internship can be summarized across both personal skill development and tangible operational analyses as follows:

- **Technical Skills Development:** Through my internship, I was able to significantly develop my technical skills in advanced production processes and strict quality control procedures. I gained valuable experience in using statistical and analytical tools to identify trends and process bottlenecks. This technical expertise will be highly valuable in my future academic and professional pursuits.
- **Soft Skills Development:** Throughout the internship, I had the continuous opportunity to develop my soft skills, such as technical communication, teamwork, and critical problem-solving under pressure. I worked closely with professionals from different departments, which allowed me to improve my cross-functional collaboration. I also had the opportunity to work on real-world efficiency problems, which helped me develop my critical thinking skills.
- **Exposure to Industry Trends:** The internship allowed me to gain direct exposure to current industry trends and real-world developments that may not always be covered in a standard classroom environment. I was able to observe the company's precise processes and learn firsthand about the economic and technical challenges facing the modern manufacturing industry.
- **Expanded Professional Network:** The internship experience provided me with the unique opportunity to network with experienced professionals from different departments, including plant managers, QC inspectors, and CNC programmers. These professional connections can potentially open doors to future career opportunities, technical mentorships, or industry collaborations.
- **Strengthened Soft Skills:** The internship experience has also allowed me to significantly strengthen my soft skills, such as inter-departmental teamwork, clear technical communication on a noisy shop floor, and rigorous time management. These skills are invaluable in any engineering profession, and their continuous development will benefit me throughout my entire career.

Conclusion

The experience at Ascent Enterprises served as a definitive bridge between theoretical mechanical engineering principles and the complexities of industrial manufacturing. Immersive engagement with CNC paradigms and shop floor operations provided a comprehensive view of how integrated departments—ranging from Sales to Quality Control—function as a cohesive unit. Contributing to time-study projects was particularly impactful, as it offered a data-driven perspective on operational efficiency and the rigorous demands of the heavy manufacturing sector.

Collaborating with seasoned professionals facilitated the development of both high-level technical competencies and essential soft skills, including cross-functional teamwork and systematic problem-solving. This tenure has established a robust professional foundation and a clear understanding of industry standards. I am sincerely grateful for the mentorship and networking opportunities provided, which have significantly sharpened my focus and readiness for a professional career in Automation & Robotics engineering.

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Discussion and Recommendations for industry

Based on my comprehensive internship experience and cycle time analyses at Ascent Enterprises, I would like to offer some discussion points and technical recommendations for the industry to carefully consider:

- **Incorporate New Technologies and Automation (Robotics):** The industry is constantly evolving, and it is essential for companies to stay up-to-date with new technologies. As identified, manual loading times fluctuate heavily due to operator fatigue, acting as a primary bottleneck. There is a highly viable scope to implement robotic "Pick-and-Place" systems directly adjacent to the CNC turning centers to drastically reduce human operator fatigue and guarantee highly stable, deeply predictable cycle times.
- **IoT Integration for Predictive Maintenance:** To aggressively tackle the issue of tool-wear-induced dimensional defects and unexpected machine breakdowns, transitioning from reactive maintenance to proactive "Predictive Maintenance" is absolutely essential. The installation of IoT smart sensors onto CNC machine spindles for real-time vibration analysis would instantly alert the operator to change the tool before it results in scrapped parts.
- **Emphasize the Importance of Internship Programs:** Internship programs are a valuable tool for students to gain hands-on experience and develop their skills. Therefore, companies should continue to emphasize the importance of internship programs and encourage more students to participate in them, fostering the next generation of engineers.
- **Opportunities for Professional Development:** Companies should continuously provide opportunities for professional development to their interns and junior staff. This can be done through dedicated training sessions, workshops, and mentoring programs. These opportunities will allow interns to gain new, cutting-edge skills and knowledge that will be highly beneficial in their future pursuits.
- **Encourage Collaboration and Teamwork:** Seamless cross-departmental collaboration and teamwork are essential for achieving a company's complex production goals. Therefore, companies should actively encourage interns to work

collaboratively on shop floor projects to develop effective technical communication skills and learn exactly how to work in a high-pressure team environment.

- **Emphasize the Importance of Soft Skills:** Soft skills such as clear communication, teamwork, and rigorous time management are absolutely essential for success in any heavy industry. Therefore, companies should emphasize the importance of developing these crucial skills during the internship program alongside technical competencies.
- **Offer Employment Opportunities:** Internship programs can be an incredibly effective way to recruit top-tier new talent for the company. Therefore, companies should offer employment opportunities to interns who have performed exceptionally well during their internship program.

In conclusion, the industry should focus on providing meaningful internship programs that strongly emphasize the importance of professional development, collaboration, and soft skills. Additionally, companies should strategically incorporate new technologies (like Robotics and IoT) into their operations. By implementing these recommendations, the industry can create a robust pipeline of talented professional engineers who will heavily contribute to the industry's continuous growth and long-term success.